

DROP TABLE IF EXISTS

bu_frd.model_cohort_behavioral_final_3m_2022_2025;

CREATE TABLE

bu_frd.model_cohort_behavioral_final_3m_2022_2025

USING PARQUET

AS

WITH eligible AS (

SELECT

c.party_id,
c.observation_date,
c.observation_year,
UPPER(c.model_split) AS model_split,
c.first_alert_date,
c.label_3m,
c.cohort_role

FROM

bu_frd.model_cohort_3m_2022_2025_v2 c

INNER JOIN

bu_frd.model_trade_coverage_13m_2022_2025_v2 v

ON c.party_id = v.party_id
AND c.observation_date = v.observation_date

WHERE

v.complete_13m_flag = 1

),

case_counts AS (

SELECT

observation_date,
COUNT(*) AS case_count

FROM eligible

WHERE label_3m = 1

GROUP BY observation_date

),

ranked_controls AS (

SELECT

```

e.*,
cc.case_count,

ROW_NUMBER() OVER (

    PARTITION BY e.observation_date

    ORDER BY
        PMOD(
            HASH(
                CONCAT(
                    CAST(e.party_id AS STRING),
                    '|',
                    CAST(e.observation_date AS STRING),
                    '|FINAL_CONTROL_10'
                )
            ),
            2147483647
        ),
        e.party_id

    ) AS control_rank

FROM eligible e

INNER JOIN case_counts cc
    ON e.observation_date = cc.observation_date

WHERE e.label_3m = 0
)

/* =====
KEEP ALL ELIGIBLE CASES
===== */

SELECT
    party_id,
    observation_date,
    observation_year,
    model_split,
    first_alert_date,
    label_3m,
    cohort_role

FROM eligible

WHERE label_3m = 1

```

UNION ALL

```
/* =====  
KEEP EXACTLY 10 CONTROLS / CASE  
===== */
```

```
SELECT  
  party_id,  
  observation_date,  
  observation_year,  
  model_split,  
  first_alert_date,  
  label_3m,  
  cohort_role
```

FROM ranked_controls

WHERE control_rank <= case_count * 10;

```
SELECT  
  model_split,  
  cohort_role,  
  label_3m,  
  
  COUNT(*) AS rows,  
  COUNT(DISTINCT party_id) AS parties
```

```
FROM  
  bu_frd.model_cohort_behavioral_final_3m_2022_2025
```

```
GROUP BY  
  model_split,  
  cohort_role,  
  label_3m
```

```
ORDER BY  
  model_split,  
  label_3m DESC;
```

```
SELECT  
  COUNT(*) AS total_rows,  
  COUNT(DISTINCT party_id) AS total_parties,  
  
  SUM(label_3m) AS cases,
```

```

SUM(
  CASE
    WHEN label_3m = 0
    THEN 1 ELSE 0
  END
) AS controls,

ROUND(
  100.0 * SUM(label_3m) / COUNT(*),
  4
) AS modeling_positive_rate_pct

FROM
  bu_frd.model_cohort_behavioral_final_3m_2022_2025;

SELECT
  observation_date,

  SUM(
    CASE WHEN label_3m = 1
    THEN 1 ELSE 0 END
  ) AS cases,

  SUM(
    CASE WHEN label_3m = 0
    THEN 1 ELSE 0 END
  ) AS controls,

  ROUND(
    1.0 *
    SUM(
      CASE WHEN label_3m = 0
      THEN 1 ELSE 0 END
    )
    /
    SUM(
      CASE WHEN label_3m = 1
      THEN 1 ELSE 0 END
    ),
    2
  ) AS control_case_ratio

FROM
  bu_frd.model_cohort_behavioral_final_3m_2022_2025

GROUP BY observation_date

```

ORDER BY observation_date;

DROP TABLE IF EXISTS

bu_frd.model_trades_behavioral_final_13m_2022_2025;

CREATE TABLE

bu_frd.model_trades_behavioral_final_13m_2022_2025

USING PARQUET

AS

SELECT

t.*

FROM

bu_frd.model_trades_13m_2022_2025_v2 t

INNER JOIN

bu_frd.model_cohort_behavioral_final_3m_2022_2025 c

ON t.party_id = c.party_id

AND t.observation_date = c.observation_date;

SELECT

COUNT(*) AS trade_rows,

COUNT(DISTINCT party_id)

AS parties,

COUNT(DISTINCT acct_id)

AS accounts,

MIN(data_date)

AS min_data_date,

MAX(data_date)

AS max_data_date

FROM

bu_frd.model_trades_behavioral_final_13m_2022_2025;

```

WITH x AS (

    SELECT
        party_id,
        observation_date,

        COUNT(DISTINCT feature_month)
        AS months_with_trade

    FROM
        bu_frd.model_trades_behavioral_final_13m_2022_2025

    GROUP BY
        party_id,
        observation_date
)

SELECT
    MIN(months_with_trade)
    AS min_months,

    percentile_approx(
        months_with_trade,
        0.50
    ) AS median_months,

    MAX(months_with_trade)
    AS max_months,

    SUM(
        CASE
            WHEN months_with_trade <> 13
            THEN 1 ELSE 0
        END
    ) AS parties_not_13_months

FROM x;

```